

WAKAYAMA, Japan

WMO: 477770

Lat: 34.2289N

Lon: 135.1642E

Elev: 60

StdP: 14.66

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	32.8	34.4	17.5	13.3	39.7	19.7	14.8	40.7	27.3	42.6	24.7	43.2	9.0	70	0.450

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.1	92.5	76.8	90.5	76.8	88.8	76.6	79.8	87.8	78.9	86.9	78.1	86.0	10.4	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.5	143.6	84.7	76.6	138.8	84.3	75.7	134.7	83.7	43.4	88.1	42.5	87.0	41.7	86.2	83.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
24.3	20.5	17.9	DB	29.7	96.4	1.2	2.0	28.8	97.9	28.0	99.0	27.4	100.1	26.5	101.6
			WB	26.1	80.7	1.7	1.5	24.9	81.8	23.9	82.7	23.0	83.5	21.8	84.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	62.8	43.0	44.1	49.8	59.2	67.8	74.2	81.7	83.8	77.4	67.2	56.8	47.6
	DBStd	14.77	3.83	4.91	5.57	5.67	3.92	3.83	3.59	2.90	4.60	4.93	5.30	4.60
	HDD50	582	219	179	74	4	0	0	0	0	0	4	102	
	HDD65	2764	681	586	471	187	18	0	0	0	32	250	539	
	CDD50	5262	3	13	68	279	553	725	982	1047	821	533	209	29
	CDD65	1969	0	0	0	12	106	276	517	582	372	100	4	0
	CDH74	17855	0	0	0	26	350	1497	5505	6994	3094	382	7	0
	CDH80	6195	0	0	0	1	27	250	1985	2991	904	37	0	0
Wind	WSAvg	8.5	9.4	9.0	9.2	8.8	8.3	7.6	8.4	8.3	7.8	7.8	8.4	9.3
Precipitation	PrecAvg	57.0	1.9	2.7	3.9	4.1	5.9	7.6	6.1	4.4	7.9	5.8	4.2	2.5
	PrecMax	76.2	4.8	6.6	7.2	8.9	17.6	14.6	16.0	21.4	17.5	13.9	13.8	6.5
	PrecMin	28.3	0.2	0.8	2.0	1.5	1.4	2.6	1.8	0.3	1.3	1.3	0.6	0.2
	PrecStd	11.7	1.2	1.5	1.3	1.8	3.7	3.1	3.7	4.4	4.4	3.3	2.8	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.6	65.3	69.5	77.3	83.1	88.7	95.0	96.0	91.2	83.6	74.1	65.9
		MCWB	53.7	58.7	58.6	62.2	65.3	74.4	77.3	78.0	75.3	72.8	63.7	58.3
	2%	DB	55.4	60.3	66.1	73.6	79.8	85.0	91.8	93.4	89.0	80.5	71.1	60.9
		MCWB	48.3	53.4	57.2	61.7	64.8	72.7	76.6	77.1	75.3	69.9	62.1	53.4
	5%	DB	52.7	56.2	63.2	71.1	77.7	82.7	89.7	91.4	87.0	78.0	68.6	58.1
		MCWB	45.3	48.7	54.7	60.3	64.3	71.3	76.9	77.0	74.5	67.6	60.1	50.7
	10%	DB	50.3	52.9	60.6	68.9	75.7	80.8	87.9	89.7	85.1	75.8	66.2	55.9
		MCWB	43.3	45.5	52.2	58.9	64.1	70.9	76.7	77.1	73.7	65.7	58.4	48.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.9	59.5	62.9	68.5	72.7	77.2	80.6	81.5	79.3	75.6	67.9	60.9
		MCDB	58.5	63.8	66.3	72.7	77.8	83.4	88.2	89.0	86.7	80.9	71.6	64.1
	2%	WB	49.8	54.5	58.8	65.5	70.4	75.9	79.4	80.2	77.8	72.5	64.3	55.0
		MCDB	53.4	59.5	64.1	70.0	75.2	81.8	87.3	88.4	85.1	78.5	68.5	59.6
	5%	WB	46.4	50.1	56.0	62.5	68.2	74.7	78.5	79.4	76.7	69.9	61.9	51.9
		MCDB	51.4	54.5	62.0	68.8	73.7	80.6	86.5	87.4	83.9	75.8	66.9	56.6
	10%	WB	43.9	46.5	53.0	60.4	66.3	73.4	77.7	78.7	75.6	67.4	59.4	49.4
		MCDB	49.6	51.9	59.7	67.2	73.1	79.0	85.5	86.6	82.7	73.8	65.1	55.0
Mean Daily Temperature Range	5% DB	MDBR	11.4	12.5	14.2	15.0	14.2	11.5	11.0	12.1	12.3	13.0	13.1	12.0
		MCDBR	15.9	17.5	17.9	17.1	17.0	13.8	13.2	14.1	13.9	14.9	15.5	16.1
		MCWBR	11.4	13.2	13.3	10.0	8.5	6.1	5.0	5.4	6.0	8.0	10.4	11.1
	5% WB	MCDBR	14.3	15.8	16.4	13.6	12.6	10.2	11.6	12.2	12.1	12.3	13.2	14.2
		MCWBR	12.3	14.1	14.6	10.4	9.3	5.5	5.1	5.4	6.3	8.5	11.1	12.0
Clear-Sky Solar Irradiance	taub	0.393	0.437	0.496	0.529	0.534	0.550	0.553	0.523	0.485	0.447	0.415	0.388	
	taud	2.203	2.092	1.945	1.915	1.976	2.013	2.052	2.131	2.208	2.230	2.226	2.230	
	Ebn at Noon	251	252	246	245	245	240	239	244	247	246	241	244	
	Edh at Noon	36	45	56	61	58	56	53	49	43	39	35	33	
All-Sky Solar Radiation	RadAvg	778	975	1284	1532	1655	1471	1620	1664	1303	1053	817	702	
	RadStd	56	73	94	144	169	146	198	143	110	106	71	43	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page